

Docker Complete Cheat Sheet with Descriptions

1. IMAGES COMMANDS

Command: `docker images`

Where to Use: Used to list all Docker images available locally.

Command: `docker pull`

Where to Use: Used to download images from Docker Hub.

Command: `docker rmi`

Where to Use: Used to delete Docker images.

Command: `docker image prune`

Where to Use: Used to remove unused images and free space.

Command: `docker build -t : .`

Where to Use: Used to build Docker images from Dockerfile.

2. CONTAINER COMMANDS

Command: `docker ps`

Where to Use: Shows running containers.

Command: `docker ps -a`

Where to Use: Shows all containers including stopped ones.

Command: `docker run`

Where to Use: Creates and starts a new container.

Command: `docker run -d`

Where to Use: Runs container in background mode.

Command: `docker run -p 5000:5000`

Where to Use: Maps container ports to host machine.

Command: `docker stop`

Where to Use: Stops running container.

Command: `docker start`

Where to Use: Starts stopped container.

Command: `docker rm`

Where to Use: Deletes container.

3. TROUBLESHOOTING COMMANDS

Command: `docker logs`

Where to Use: Used to view container logs.

Command: `docker logs -f`

Where to Use: Used to monitor logs live.

Command: `docker exec -it bash`

Where to Use: Opens terminal inside container.

Command: `docker stats`

Where to Use: Displays CPU and memory usage of containers.

4. DOCKER HUB COMMANDS

Command: `docker login`

Where to Use: Used to login into Docker Hub.

Command: `docker push /`

Where to Use: Uploads image to Docker Hub.

Command: `docker search`

Where to Use: Search images from Docker Hub.

5. VOLUME COMMANDS

Command: `docker volume ls`

Where to Use: Lists all Docker volumes.

Command: `docker volume create`

Where to Use: Creates persistent storage volume.

Command: `docker run -v :/data`

Where to Use: Mounts volume to container.

6. NETWORK COMMANDS

Command: `docker network ls`

Where to Use: Lists Docker networks.

Command: `docker network create`

Where to Use: Creates custom network.

Command: `docker network connect`

Where to Use: Connects container to network.

7. DOCKER COMPOSE COMMANDS

Command: `docker compose up`

Where to Use: Starts multi-container applications.

Command: `docker compose up -d`

Where to Use: Runs services in background.

Command: `docker compose down`

Where to Use: Stops and removes compose services.

8. CLEANUP COMMANDS

Command: `docker system prune -a`

Where to Use: Removes unused containers, images, networks, and cache.

Command: `docker volume prune`

Where to Use: Removes unused volumes.

9. SAVE & LOAD COMMANDS

Command: `docker save -o myimage.tar`

Where to Use: Saves Docker image into tar file.

Command: `docker load -i myimage.tar`

Where to Use: Loads image from tar file.

10. SYSTEM COMMANDS

Command: `docker --version`

Where to Use: Checks installed Docker version.

Command: `docker info`

Where to Use: Displays Docker system information.

Command: `docker system df`

Where to Use: Shows Docker disk usage.